VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Digital Payload Processor (DPP) MPU

AFLCMC/WA

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1.0 Overview

A Digital Payload Processor (DPP) is a Level 2 Minimum Procurable Unit (MPU) in the Agile Mission Suite (AMS) Government Reference Architecture (GRA). DPPs provide processing for the Mission Processing Subsystem (MPS), Multi-Function Processor (MFP), and the Multi-Function Aperture (MFA) layers. DPP MPUs are housed in Processor Chassis and Backplanes (PCaBs) and are a part of the Compute Cluster Farm (CCF).

1.1 Purpose

The DPP MPU Volume introduces the DPP MPU and describes its key features, logical context, interfaces, and compliance requirements. The key features (**Table 2.1.1**) provide design guidance on what is typically expected for the DPP MPU. The interface diagram contains a graphical representation of the types of interfaces that can exist (from the perspective of the DPP MPU). The Interface Table (**Table 2.3.1**) contains a textual representation of the interfaces and their descriptions. Compliance requirements (**Table 3.1.1**) describe what is required for the DPP MPU to achieve an AMS GRA compliance tier.

1.2 Scope

The DPP Volume supports compliance assessment for AMS GRA adopters.

1.3 Authoritative References

Authoritative references are key documents that a procuring authority, Mission System Integrator (MSI), or MPU provider refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. An AMS GRA entity is compliant when it provides the required prescriptive constructs (e.g. interfaces for an MPU). See **Table 1.3.1** for a list of authoritative references for the DPP MPU Volume.

Standard	Title	Version/Date
AMS GRA Models	AMS GRA Architecture Model	14.0
AMS GRA Volumes	Compliance Assessment Approach Volume	14.0
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	Software Architecture Volume	14.0
AMS GRA Volumes	DPP Compliance Documentation Template	14.0

Standard	Title	Version/Date
IEEE 1588-2008	IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems	2008-07-24

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing a Compliance Assessment of the DPP MPU.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the DPP MPU Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Time Distribution Volume	14.0
AMS GRA Volumes	PCaB MPU Volume	14.0

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the DPP MPU and Related Entities.

2.0 Digital Payload Processor (DPP)

A DPP MPU is a hardware module that provides computing resources and is physically hosted within a PCaB. The primary purpose of a DPP is to host MFP and MPS Layer software Services on one or more processing components contained in the DPP (e.g. General Purpose Processor (GPP) components, Field-Programmable Gate Array (FPGA) components, and Graphics Processing Unit (GPU) components). A DPP may contain any mixture of one or more of these processing components. AMS GRA does not prescribe specific processing capacity of these processing components so that DPP providers may customize DPP designs for different uses, such as MFP Layer signal processing or MPS layer command and control software. To consider a component a processing component within the DPP, the DPP Development Kit (DK) will have to enable Service software to run on that component. Likewise, AMS GRA refers to Service source code and images targeting processing components to be software, regardless of the processing component type. The different processing components each host the following types of software:

- GPP components host traditional Central Processing Unit (CPU)-based Service software. GPP components typically host logic for command and control, networking interfaces, and complicated message handling. GPP components can also host more demanding signal processing depending on features such as clock rates and multicore processing.
- FPGA components host software that typically implements special interface handling, or computationally intensive software logic for high data rate or low latency applications. FPGA components may be high capacity FPGA System on a Chip (SoC) components, but they may also be simpler Complex Programmable Logic Device (CPLD) components or other similar devices as long as the DPP allows hosting of Service software on the components through its DK.
- GPU components host software that typically implements highly parallelized algorithms that take advantage of the high number of processing cores provided by GPUs.

DPPs are not limited to containing a single type of processing component and could contain all of the processing components defined in this chapter.

There are a number of interfaces optionally available to a DPP MPU. The Abstract Service Bus (ASB) provides an external and/or internal communication interface between DPPs supporting Open Mission Systems (OMS) communication. The Avionics Backplane Bus (ABB) provides another external and/or internal communication interface between DPPs.

Several aspects of a DPP affect its operational use. These aspects divide into three categories of elements: Unchangeable, Field-Loadable, and Maintenance-Loadable. The following paragraphs provide more detail. Note

that the modification (i.e., update and reload) of any software Loadable element must not change the DPP part number. This versioning approach avoids the logistic issues associated with configuration changes of the hardware part numbers.

Unchangeable elements are items that cannot change after delivery of a module, either with or without special test equipment. Typically, unchangeable elements are *Write Once* values within a processor/Application Specific Integrated Cir t (ASIC). Programming of the *Write Once* values occurs at the processor/ASIC factory or the DPP Original Equipment Manufacturer (OEM). However, a DPP vendor could consider some very early stages of the boot software as unchangeable.

A DPP MPU may have elements/components that may be loaded/updated in the field. Field-Loadable elements contain software, firmware, and/or parametric data that may be loaded without special test equipment in a deployed platform (i.e., the DPP MPU does not need removal from its PCaB). Examples of items considered Field-Loadable elements are DPP Non-Volatile Memory (NVM), containing boot software or configuration data files.

A DPP MPU may also have elements/components that must be loaded/updated with special test equipment. These elements/components are Maintenance-Loadable elements. The special test equipment required to load these components may or may not require removal of the module from its PCaB. Examples of Maintenance-Loadable elements are CPLD images, power sequencer logic, and early stages of boot software.

2.1 Key Features

Table 2.1.1 identifies the key features of DPPs and the benefit of these features. This table is not prescriptive but identifies a number of attributes that AMS GRA definitions and interfaces are designed to support. This table is included in the DPP documentation template so that DPP providers can summarize the specific features supported by a DPP instantiation.

Key Feature	Benefit
Communication	The DPP allows the implementation of communications, which serves as a means of transferring data or establishing connectivity between MPUs.
Memory	Memory provides for fast volatile storage supporting runtime software execution, (e.g., RAM).
NVM	NVM optionally provides for localized persistent data storage, e.g., offline storage of configured software binaries and configuration files, and offline storage of boot and firmware images. The majority of persistent data storage resides within the MLS/NVM Data Storage MPU. Ideally, only data material is stored in the DPP MPU NVM such that the DPP MPU remains at rest.

Key Feature	Benefit
Processing	DPPs contain one or more Processing Components of type GPP Component, FPGA Component and GPU Component.
Security	The security feature optionally provides localized hardware-based security functions and processing (e.g., a TPM) notionally coordinated with security functions and processing outside of the DPP.

Table 2.1.1: The DPP Key Features Table Describes the Specific Features Supported by a DPP Instantiation.

2.2 Interface Diagram

A DPP consists of one or more hardware modules that plug into a PCaB. The interface diagram only contains logical entities.

Figure 2.2.1 contains a graphic that depicts the AMS GRA Service and its execution environment provided by the DPP MPU logical interfaces. Interface connections denoted as pass-through only represent the situation where the sending MPU is not the original source. Pass-through interfaces do not use or modify what is provided to the receiving MPU.

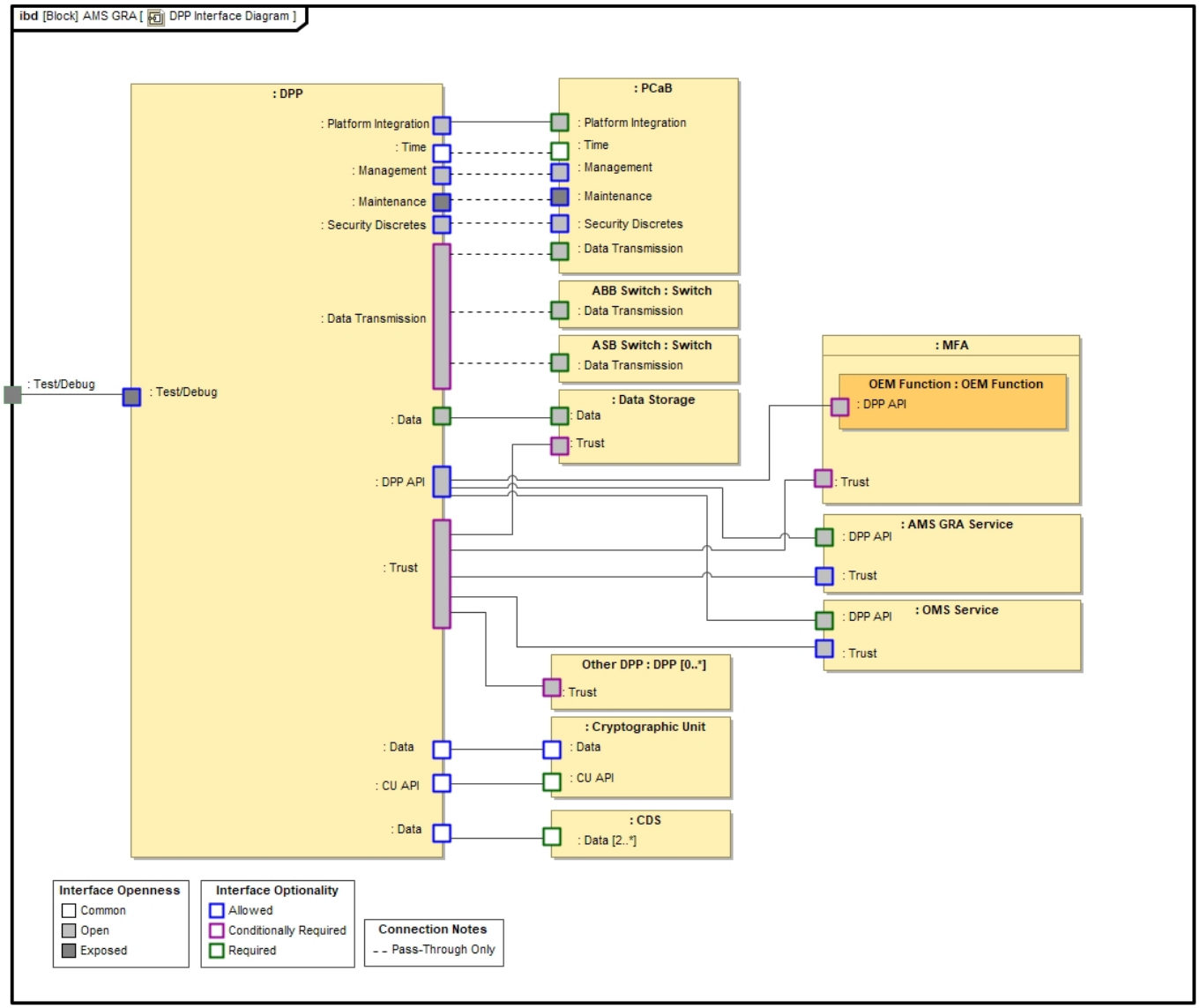


Figure 2.2.1: This DPP MPU Interface Diagram Depicts how a DPP Interfaces with Other AMS GRA MPUs.

Figure 2.2.2 contains a graphic that depicts the DPP Processing Diagram. The DPP is composed of one or more processing components. Each processing component in turn is generalized into a GPP Component, a GPU Component, or an FPGA Component.

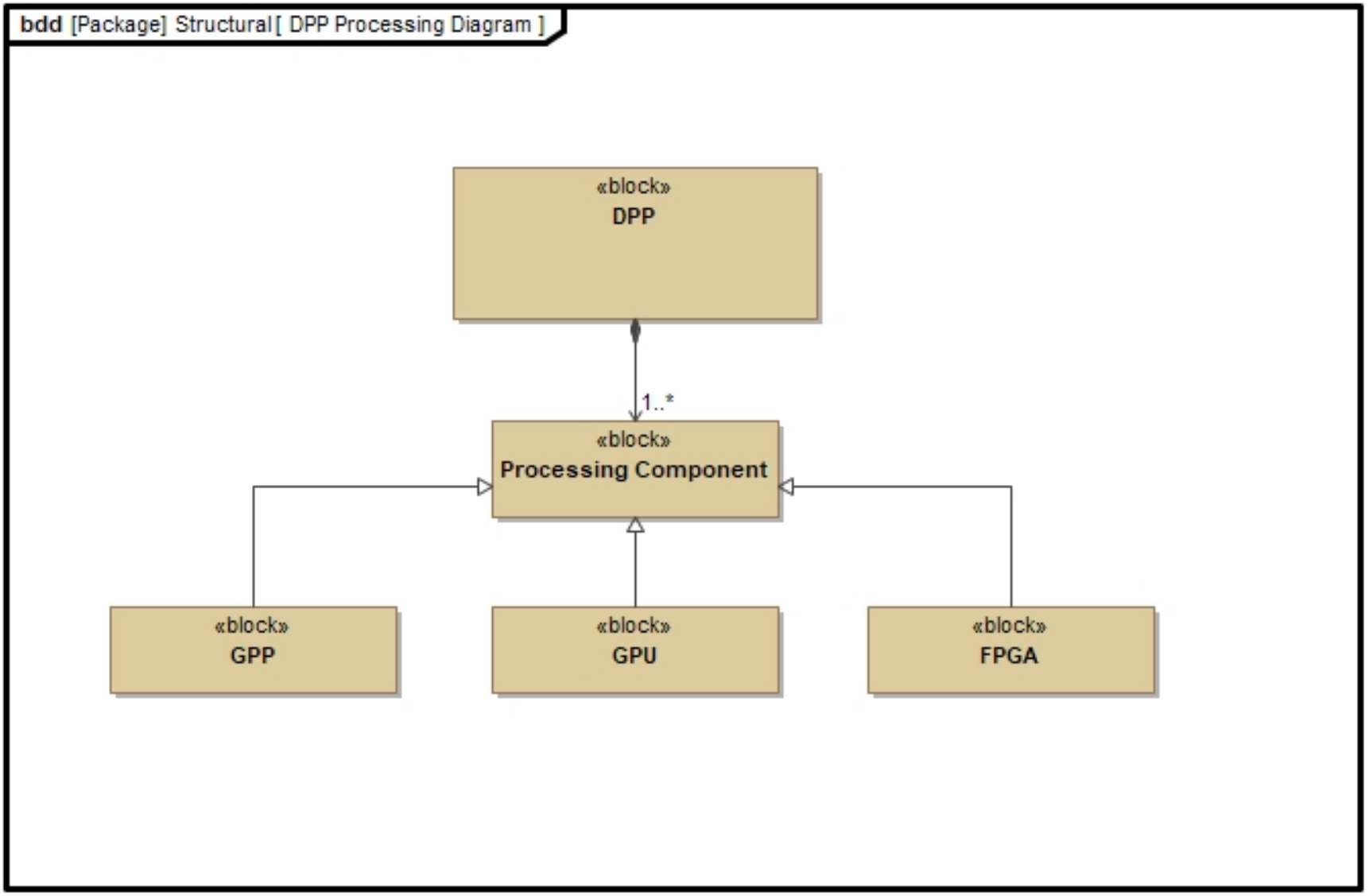


Figure 2.2.2: This DPP Processing Diagram Depicts the Components of a DPP and the Forms Those Components Can Take.

2.3 Interface Table

Table 2.3.1 identifies the DPP interfaces depicted in **Figure 2.2.1**. When determining whether a given MPU or infrastructure is AMS GRA compliant, interface optionality is defined as follows:

- Required interfaces must be implemented.
- Conditionally Required interfaces must be implemented when specific conditions are satisfied.
- Allowed interfaces are optional to implement.

Interface	Documentation	Interface Optionality	Condition
Data	The Data interface specifies the data interfaces between the DPP and other MPUs.	Required	N/A
Data Transmission	Data Transmission represents the interface support used to facilitate a data transmission connection. This interface is not concerned with what type of data is being transmitted, it only denotes what MPUs can be connected for transmitting data.	Conditionally Required	Required when the DPP connects to other MPUs and transfers data.
DPP API	The DPP Application Programming Interface (API) provides Level 1 MPUs with access to various facilities provided by the DPP Compute Environment.	Allowed	N/A
Maintenance	The Maintenance interface provides the capability to perform preventative and corrective maintenance while installed on the platform.	Allowed	N/A
Management	The Management interface provides the capability to support management of the MPU in an AMS GRA compliant mission system. Example management for the DPP includes centralized control and monitoring (e.g. IPMI).	Allowed	N/A
Platform Integration	The Platform Integration interface represents the general physical interfaces that an MPU will typically have with the host platform, including but not limited to mounting, electrical, hydraulic, and cooling connections.	Allowed	N/A
Security Discretes	The Security Discretes interface specifies the security interfaces between the DPP and the platform.	Allowed	N/A
Data	The Data interface between the DPP and the CU provides connected MPUs with access to Cryptography functionality.	Allowed	N/A

Interface	Documentation	Interface Optionality	Condition
CU API	The CU API enables AMS GRA Service to access the CU's cryptographic functionalities	Allowed	N/A
Time	The Time Distribution interface specifies the interface between the DPP and the platform.	Allowed	N/A
Trust	The Trust interface enables secure connectivity between endpoints using cryptographic means to establish a secure channel. This channel may be used to verify integrity, authenticate endpoints, or to transfer data requiring higher levels of assurance than normal data transfers, such as CPI, or Software. The implementation of this interface may vary depending on assurance level, exportability, and the various mission specific requirements of the adopting program.	Conditionally Required	Required if the MPU is used to transfer mission data, software, or associated Key Material, and there are no other interfaces implemented which support integrity verification.
Test/Debug	The Test/Debug interface specifies an interface to use for test equipment purposes between the DPP and Platform Test/Debug.	Allowed	N/A

Table 2.3.1: The Procuring Authority, MPU Provider and MSI Use this Table to Identify the Interfaces to Implement for a Compliant MPU or Infrastructure.

3.0 Compliance Requirements

3.1 DPP Requirements

Table 3.1.1 states the applicable MPU compliance requirements for the DPP MPU.

See the Compliance Assessment Approach Volume for more information on the tiered requirement strategy as well as the definitions of the interface characteristics.

See the Mission System Volume for additional details about MPU Fault Reporting.

MPU developers coordinate with the procuring authority and applicable MSIs to receive additional guidance beyond basic compliance. The additional guidance may include architectural guidance to ensure the MPU integrates within the overall Mission System architecture as well as guidance to align with Commercial-Off-The-Shelf (COTS) standards such as physical form factor, cooling strategy, and environmental operating conditions.

See **Table 3.2.1** for the relevant AMS GRA DK requirements.

Note: The verification methods listed in the requirements table are AMS GRA recommendations. An adopting program may alter the recommended verification methods In Accordance With (IAW) the needs of the program. Refer to the Compliance Assessment Approach Volume for more information.

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-001	An MPU shall be documented IAW the corresponding AMS GRA compliance documentation template or contract template, as applicable.	Tier 1	Inspection	
GRA-MPU-011	An MPU that hosts Level 1 MPUs shall provide a DK IAW DK requirements.	Tier 1	Inspection	
GRA-MPU-014	An MPU shall implement a PTP client compatible with IEEE 1588-2008.	Tier 1	Inspection	
GRA-MPU-024	An MPU that interacts with the CDS or Cryptographic Unit shall use the CDS IDD or CU IDD respectively interface for all exchanges with the Cryptographic Unit or CDS.	Tier 1	Inspection	
GRA-MPU-025	An MPU shall report faults with the associated fault status.	Tier 1	Inspection	Off-nominal behavior detailed in compliance documentation.
GRA-MPU-026	An MPU shall implement Required interfaces specified IAW the AMS GRA Model.	Tier 1	Inspection	
GRA-MPU-027	An MPU shall implement all interfaces with the level of Openness specified IAW the AMS GRA Model.	Tier 1	Analysis	
GRA-MPU-029	An MPU shall implement only the interfaces in the corresponding MPU interface table.	Tier 2	Inspection	
GRA-MPU-030	An MPU shall document interface openness IAW the Compliance Assessment Approach.	Tier 1	Inspection	
GRA-MPU-031	An MPU shall include a model compatible with MBSE content contained in the AMS GRA SK release with which the MPU is compliant.	Tier 2	Inspection	
GRA-MPU-032	An MPU shall include a model that identifies all external MPU interfaces.	Tier 2	Inspection	
GRA-MPU-038	An MPU shall use open, commercial standards for the Management interface.	Tier 1	Inspection	
GRA-MPU-041	An MPU shall document SAST results for all microprocessor software source code in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	
GRA-MPU-042	An MPU shall document DAST results for all microprocessor software in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-046	An MPU shall provide automated tests of non-COTS microprocessor software with at least 50% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 1	Demonstration	Only applicable to MPUs that include non-COTS microprocessor code.
GRA-MPU-058	An MPU shall encrypt classified DaR using NSA-approved algorithms.	Tier 1	Analysis	
GRA-MPU-059	An MPU that decrypts and/or encrypts classified data and/or CPI shall only store classified plain text keys in volatile memory.	Tier 1	Inspection	
GRA-MPU-060	An MPU's write-protected memory shall be modifiable through a hardware element in addition to software.	Tier 1	Analysis	
GRA-MPU-061	An MPU shall ensure the authenticity and integrity of MPU software and firmware during MPU initialization.	Tier 1	Inspection	
GRA-MPU-062	An MPU shall support integrity verification to the Mission System over a secure communication channel.	Tier 1	Demonstration	
GRA-MPU-063	An MPU shall implement secure communications using open standards.	Tier 1	Analysis	
GRA-MPU-064	An MPU shall use NSA-approved algorithms to authenticate and verify the integrity of a software load when received over a non-secure communication channel.	Tier 1	Demonstration	
GRA-MPU-065	An MPU shall use NSA-approved algorithms to authenticate and verify the integrity of software locally stored at rest, not including bootloaders; prior to updating, loading, and executing.	Tier 1	Inspection	
GRA-MPU-066	An MPU shall support the use of a Mission System-provided RoT.	Tier 1	Demonstration	
GRA-MPU-067	An MPU shall publish details of security relevant events in an open format.	Tier 1	Demonstration	
GRA-MPU-068	An MPU shall boot from write-protected memory.	Tier 1	Demonstration	
GRA-MPU-077	An MPU shall provide automated tests of non-COTS microprocessor software with at least 80% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 3	Demonstration	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-078	An MPU model shall be capable of automated verification with AMS GRA models.	Tier 3	Inspection	
GRA-MPU-079	An MPU shall include a model that provides definition of all external MPU interfaces.	Tier 3	Inspection	
GRA-MPU-085	An MPU shall report faults with the associated fault status using the applicable interface. (See notes).	Tier 2	Inspection	Cryptographic Unit MPUs and CDS MPUs report faults as defined in the CDS and CU IDD. Local Function MPUs report faults as defined in Local Function DK. MFA MPUs report faults as defined in the applicable MEL IDD. AMS GRA Service, CMM, DPP, Data Storage, Switch, OMS Service, and PCaB MPUs report faults using a software interface in UCI format.
GRA-MPU-088	An MPU shall report health and status on its Management Interface.	Tier 1	Demonstration	
GRA-MPU-091	An MPU shall report BIT information on its Management Interface.	Tier 1	Inspection	Note: MFA reports BIT information via the MEL interface
GRA-MPU-092	An MPU shall include a software bill of materials in an open and non-proprietary format.	Tier 2	Inspection	

Table 3.1.1: A DPP's Adherence to the Compliance Requirements Determines Its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

3.2 Development Kit (DK) Requirements

Table 3.2.1 provides DK requirements that apply to DPP MPUs. The table indicates the processor types (e.g., GPP, GPU, and FPGA) to which each requirement applies in order to assist the respective DK definition activity. In some cases, the MSI may replace the content in a requirement originally provided by the DPP MPU provider. The column for MSI indicates which requirements the MSI may serve as an alternate provider of the content. There are three patterns that refine some of the DK requirement definitions in **Table 3.2.1**.

First, some requirements exist in pairs such that one requirement discusses a need for a commodity item while the other requirement presents the need for a custom tool.

Second, and related to the first, some requirements only request the identification of certain DK content while others require the actual provision of the content. For example, a DK provider that uses a commodity item with a license would identify the item so a DK user could procure the item and acquire a license. A DK provider that uses a custom item would need to provide the item to the DK user.

Third, some requirements include the phrase “if required for development." These requirements may or may not apply to the compliance assessment, depending on the DK provider’s implementation of the DK. If the DK implementation enables development without the need for a requirement that is marked in this way, then the requirement does not apply.

For the tier definitions of some DK requirements, AMS GRA applies a versioning philosophy. Tier 3 compliance requires the identification of a specific version of a standard (e.g., hardware form factor, programming language, and network protocol) or development tool, provided the standard or tool is backward compatible. The identification of specific versions occurs to support testing purposes or to support a lack of backward compatibility of features used by AMS GRA. AMS GRA instantiations should enable version updates that eliminate reliance on deprecated or removed features in obsolete versions of standards or tools.

See the Software Architecture Volume for more information regarding DKs and the overall software development environment.

Note: The verification methods listed in the requirements table are AMS GRA recommendations. An adopting program may alter the recommended verification methods IAW the needs of the program. Refer to the Compliance Assessment Approach Volume for more information.

Id	Text	Compli-ance Tier	Recommended Verification Method	GPP	GPU	FPGA	MSI	Notes
GRA-DK-001	A DK shall provide a developer guide and support documentation, including a Software Bill of Materials, for a developer to use the DK and integrate within a DevSecOps tool chain.	Tier 1	Inspection	Yes	Yes	Yes	No	
GRA-DK-002	A DK shall include a build environment with pre-installed build infrastructure (where licensing/redistribution restrictions allow) if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	Applies only to DPP Build Infrastructure could consist of virtual machines and/or containers
GRA-DK-003	A DK shall include a development environment with libraries and tools pre-installed (where licensing/redistribution restrictions allow) if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	Development environment could consist of virtual machines or containers
GRA-DK-004	A DK shall provide simulators if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-005	A DK shall provide emulators if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-006	A DK shall provide software configuration management support to facilitate integration of software into the DevSecOps environment.	Tier 1	Inspection	Yes	Yes	Yes	Yes	

Id	Text	Compli-ance Tier	Recommended Verification Method	GPP	GPU	FPGA	MSI	Notes
GRA-DK-007	A DK shall provide build infrastructure configuration to configure the build environment.	Tier 1	Inspection	Yes	Yes	No	Yes	Infrastructure configuration could be provided either as documentation or as code Updated to "build infrastructure" and "build environment". The term "build environment" was used to allow developers to define their development workspace.
GRA-DK-008	A DK shall provide a Service Framework Reference Implementation for the design and implementation of a Skill.	Tier 3	Inspection	Yes	Yes	Yes	No	
GRA-DK-009	A DK shall provide a container framework if required for development.	Tier 1	Inspection	Yes	No	No	Yes	
GRA-DK-010	A DK shall provide a unit testing framework for a DPP. This framework enables the creation of unit tests for Level 1 MPUs.	Tier 1	Inspection	Yes	Yes	Yes	No	
GRA-DK-011	A DK shall provide target support packages to configure the specific processing resource.	Tier 1	Inspection	Yes	Yes	Yes	No	
GRA-DK-012	A DK shall identify specific versions of commodity build support utilities in a machine-readable format.	Tier 1	Inspection	Yes	Yes	Yes	Yes	An SBOM is a means to achieve machine readability.
GRA-DK-013	A DK shall provide custom build support utilities if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-014	A DK shall provide documentation of compiler flags and/or configuration properties if required for development.	Tier 1	Inspection	Yes	Yes	No	Yes	

Id	Text	Compli-ance Tier	Recommended Verification Method	GPP	GPU	FPGA	MSI	Notes
GRA-DK-015	A DK shall provide test tools for applicable Level 1 MPUs.	Tier 1	Inspection	Yes	Yes	Yes	Yes	The level 2 DPP MPU provides a DK for applicable level 1 MPUs such as AMS or OMS services.
GRA-DK-016	A DK shall identify specific versions of commodity tools if required for development in a machine-readable format.	Tier 1	Inspection	Yes	Yes	Yes	Yes	An SBOM is a means to achieve machine readability.
GRA-DK-017	A DK shall provide custom tools if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-018	A DK shall identify software integrity tools for a DPP.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-019	A DK shall provide custom debug tools if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-020	A DK shall identify specific versions of commodity packaging scripts and/or tools in a machine-readable format.	Tier 1	Inspection	Yes	Yes	Yes	Yes	An SBOM is a means to achieve machine readability.
GRA-DK-021	A DK shall provide custom packaging scripts and/or tools if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-022	A DK shall identify DASTs if required for development.	Tier 3	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-023	A DK shall identify SASTs and associated configurations using IaC if required for development.	Tier 3	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-024	A DK shall identify specific versions of commodity release packaging in a machine-readable format.	Tier 1	Inspection	Yes	Yes	Yes	Yes	An SBOM is a means to achieve machine readability.
GRA-DK-025	A DK shall provide custom release packaging if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-026	A DK shall identify specific versions of commodity dependent documentation and/or libraries in a machine-readable format if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	An SBOM is a means to achieve machine readability.

Id	Text	Compli-ance Tier	Recommended Verification Method	GPP	GPU	FPGA	MSI	Notes
GRA-DK-027	A DK shall provide specialty infrastructure libraries if required for development.	Tier 1	Inspection	Yes	Yes	No	Yes	
GRA-DK-028	A DK shall provide MASI elements whose behavior is required for development.	Tier 1	Inspection	Yes	No	No	Yes	
GRA-DK-029	A DK shall provide the processing resource tool configuration if required for development in a machine-readable format.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-030	A DK shall provide FPGA intellectual property Blocks if required for development.	Tier 1	Inspection	No	No	Yes	Yes	
GRA-DK-032	When used for DPP development, the DK shall provide a DPP API implementation required for development.	Tier 1	Inspection	Yes	No	No	Yes	Machine readable is supposed to be an SBOM
GRA-DK-033	A DK shall provide CI/CD support configuration files necessary to integrate DK components into a DevSecOps pipeline.	Tier 1	Inspection	Yes	Yes	No	Yes	
GRA-DK-034	A DK shall provide available OEM Function interfaces when used for Local Function development.	Tier 1	Inspection	Yes	Yes	Yes	No	
GRA-DK-035	A DK shall include a containerized build environment with pre-installed build infrastructure (where licensing/redistribution restrictions allow) if required for development.	Tier 2	Inspection	Yes	Yes	Yes	Yes	Applies only to DPP Build Infrastructure must consist of containers only
GRA-DK-036	A DK shall include a containerized build environment with pre-installed build infrastructure (where licensing/redistribution restrictions allow) if required for development.	Tier 3	Inspection	Yes	Yes	Yes	Yes	Applies only to MFA Build Infrastructure must consist of containers only
GRA-DK-037	A DK development environment shall include debugging tools and libraries created by the MPU provider to streamline development for the target MPU.	Tier 3	Inspection	Yes	No	No	No	
GRA-DK-038	A DK shall provide build infrastructure configuration as IaC to configure the build environment.	Tier 3	Inspection	Yes	Yes	No	Yes	

Id	Text	Compli-ance Tier	Recommended Verification Method	GPP	GPU	FPGA	MSI	Notes
GRA-DK-039	A DK shall provide a unit testing framework for an MFA. This framework enables the creation of unit tests for Level 1 MPUs.	Tier 3	Inspection	Yes	Yes	Yes	No	
GRA-DK-040	A DK shall provide documentation (as `Cmake` or equivalent project settings as code) of compiler flags and/or configuration properties if required for development.	Tier 2	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-041	A DK shall document all licensed/non-redistributable tools and libraries required for the development environment.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-042	A DK shall document all licensed/non-redistributable software required for the build environment.	Tier 1	Inspection	Yes	Yes	Yes	Yes	Applies only to DPP
GRA-DK-043	A DK shall document all licensed/non-redistributable software required for the build environment.	Tier 3	Inspection	Yes	Yes	Yes	Yes	Applies only to MFA
GRA-DK-044	A DK shall identify specific versions of commodity packaging scripts and/or tools.	Tier 3	Inspection	Yes	Yes	Yes	Yes	Applies only to MFA

Table 3.2.1: A DK's Adherence to the Compliance Requirements Determines Its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
API	Application Programming Interface
ASB	Abstract Service Bus
ASIC	Application Specific Integrated Cir t
BIT	Built-In Test
CCF	Compute Cluster Farm
CDS	Cross Domain Solution
CI/CD	Continuous Integration/Continuous Delivery
CMM	Countermeasures Manager
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
CPLD	Complex Programmable Logic Device
CPU	Central Processing Unit
CU	Cryptographic Unit
DaR	Data at Rest
DAST	Dynamic Application Security Testing
DevSecOps	Development, Security, and Operations
DK	Development Kit
DPP	Digital Payload Processor
FPGA	Field-Programmable Gate Array
GPP	General Purpose Processor
GPU	Graphics Processing Unit
GRA	Government Reference Architecture
IaC	Infrastructure as Code
IAW	In Accordance With
IDD	Interface Description Document

IEEE	The Institute of Electrical and Electronics Engineers
IPMI	Intelligent Platform Management Interface
MASI	Mission Agnostic Service Infrastructure
MBSE	Model-Based Systems Engineering
MEL	MFA Encapsulation Layer
MFA	Multi-Function Aperture
MFP	Multi-Function Processor
MLS	Multi-Level Security
MPS	Mission Processing Subsystem
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NSA	National Security Agency
NVM	Non-Volatile Memory
OEM	Original Equipment Manufacturer
OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
PTP	Precision Time Protocol
RAM	Random Access Memory
RoT	Root of Trust
SAST	Static Application Security Testing
SBOM	Software Bill of Materials
SK	Starter Kit
SoC	System on a Chip
TPM	Trusted Platform Module
UCI	Universal Command and Control Interface

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision adds: 1. Communication as a Key Feature within the Key Features Table 2. A fix for the DPP API Interface Documentation	July 24, 2025	SK13.0
B	This revision reflects the deprecation of Security Module content. Adds new requirements GRA-MPU-088, GRA-MPU-91, and GRA-MPU-92.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the DPP Volume Between Documentation Releases.